

The company estimates that a solar farm built using these collectors and producing a gigawatt of electricity would cover about 15 square kilometres.

Solar nanotrees to produce hydrogen fuel

The only waste product that hydrogen generates is water, making it a clean fuel. However, conventional electricity is used to generate hydrogen. Researchers at the University of California, San Diego, have developed a 3D branched nanowire array that could prove to be a clean way to generate clean fuel on a large scale.

The nanowires, made from silicon and zinc oxide, are branching structures with vertical wires that sprout countless branches. This tree-like structure maximises the amount of solar energy captured. The vertical structures absorb sunlight, while the flat surfaces reflect it.

Using this nanotree structure, the researchers produced hydrogen in photo-electrochemical water-splitting. This process usually uses planar solar cells to produce hydrogen in a process similar to the electrolysis of water, but the nanowire arrays produce more hydrogen fuel efficiently.

The researchers are also looking to use the structure to mimic photosynthesis in a device that captures atmospheric carbon-dioxide to reduce carbon emissions, while producing hydrogen fuel.

Active band-stop filter allows increase in phone volume

It is now possible to get louder sound out of the tiny loudspeakers of mobile phones without overheating and damaging them. ST-Ericsson, a company that creates mobile platforms and wireless semiconductors, has developed a system that increases power supplied to the speakers while using a band-stop filter.

As power is increased, overheating and over-excursion occurs first at the resonant frequency of the loud-



Heat-assisted magnetic recording technology from Seagate

speaker. By putting a band-stop filter, also known as a notch filter, in the frequency spectrum presented to the speaker, more power can be fed in. The resonant frequency changes with temperature. So the audio processor analyses the current spectrum to locate the resonance, then shifts the notch appropriately. This allows the speaker to be driven from a 2.5W audio amplifier.

The scheme has been implemented on a Class-D audio amplifier chip called AV2001.

New technology may double the capacity of hard drives

It is now possible to create super-dense memory storage using heat-assisted magnetic recording (HAMR) technology—an advancement over perpendicular magnetic recording (PMR) being utilised in current hard drives. Seagate claims to have become the first hard-drive maker to achieve the storage density of 0.15 terabit per square centimetre (one terabit per square inch).

This breakthrough promises to double the storage capacity of today's hard drives and, subsequently, give rise to 8.9cm (3.5-inch) hard drives with an extraordinary capacity of up to 60 terabytes.

The current hard drive technology PMR is used to record digitised data on the spinning platters inside the hard drive. However, by chemically encoding an organised molecular pattern onto

the platter's substrate, HAMR can put a single bit on every magnetic grain, increasing the amount of data that can fit, and remain stable. Using HAMR technology, a linear bit density of about 787 kilobits per cm (two million bits per inch) has been achieved. This has resulted in a data density that is 55 per cent higher than today's areal density ceiling of 96 gigabits per square centimetre (620 gigabits per square inch).

Sensors to spot material defects

A team of scientists at Indira Gandhi Centre for Atomic Research has developed optical sensors for extracting defect feature information from images. The sensors can be used to detect and view structural defects in materials and components.

The magnetically polarisable sensor has oil droplets as an emulsion of water. The oil droplets about 200 nanometres in diameter contain magnetic particles about 6.6 nanometres in size. The working principle of the sensor is that defective regions in a material produce change in magnetic flux. The magnetic flux leaks if the material has any structural defect. The nanofluid inside the optical sensor immediately forms a one-dimensional array or chain along the direction of the magnetic field. While the optical sensor can provide the result immediately, the dimension of the defect can be found using certain modelling.

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